



**SIMATS ENGINEERING**



**Saveetha Institute of Medical and Technical Sciences**

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# **ASSESSMENT TOOL -1**

## **CO2\_AT1**

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**Course code :** CSA1613

**Course name:** Data Warehousing And Data Mining

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## SET 3

### 1. Data Cleaning using Duplicate Detection (Patient Registration) (5 Marks)

The following patient IDs were extracted from a registration database.

P101, P102, P105, P103, P105, P107, P108, P103, P110, P112

Perform the following:

1. Identify duplicate records.
2. Remove duplicate entries.
3. Determine the number of unique patients.

ANSWER :-

Data Cleaning using Duplicate Detection (5 Marks)

Given dataset of Patient IDs:

P101, P102, P105, P103, P105, P107, P108, P103, P110, P112

**Identify duplicate records :**

By counting the occurrences of each ID:

- P103 appears 2 times
- P105 appears 2 times

Duplicates identified: P103 and P105

**Remove duplicate entries**

Retaining only the first occurrence of each distinct ID gives:

P101, P102, P105, P103, P107, P108, P110, P112

**Determine the number of unique patients:**

Counting the remaining distinct entries:

Number of unique patients = 8

## 2. Covariance Analysis between Treatment Cost and Hospital Stay (10 Marks)

The following data represent treatment cost (₹ thousand) and hospital stay (days).

| Treatment Cost (x) | Hospital Stay (y) |
|--------------------|-------------------|
| 20                 | 3                 |
| 25                 | 4                 |
| 30                 | 5                 |
| 35                 | 6                 |
| 40                 | 8                 |

Perform the following:

1. Compute the mean of x and y.
2. Calculate the covariance.
3. Interpret the relationship between treatment cost and duration of stay.

ANSWER :-

Covariance Analysis between Treatment Cost and Hospital Stay :

Given data for  $n = 5$  patients

| Treatment Cost (x) | Hospital Stay (y) |
|--------------------|-------------------|
| 20                 | 3                 |
| 25                 | 4                 |
| 30                 | 5                 |
| 35                 | 6                 |
| 40                 | 8                 |

**Compute the mean of  $x$  and  $y$** 

$$\bar{x} = \frac{\sum x}{n} = \frac{20 + 25 + 30 + 35 + 40}{5} = \frac{150}{5} = 30$$

$$\bar{y} = \frac{\sum y}{n} = \frac{3 + 4 + 5 + 6 + 8}{5} = \frac{26}{5} = 5.2$$

**Calculate the covariance**

Using the sample covariance formula:

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

| $x_i$ | $y_i$ | $x_i - \bar{x}$ | $y_i - \bar{y}$ | $(x_i - \bar{x})(y_i - \bar{y})$ |
|-------|-------|-----------------|-----------------|----------------------------------|
| 20    | 3     | -10             | -2.2            | 22.0                             |
| 25    | 4     | -5              | -1.2            | 6.0                              |
| 30    | 5     | 0               | -0.2            | 0.0                              |
| 35    | 6     | 5               | 0.8             | 4.0                              |
| 40    | 8     | 10              | 2.8             | 28.0                             |

**Sample Covariance:**

$$\text{Cov}(x, y) = \frac{60.0}{5 - 1} = \frac{60.0}{4} = 15$$

(Note: If using population covariance formula with  $n = 5$ ,  $\text{Cov}(x, y) = \frac{60}{5} = 12$ )

**Interpret the relationship**

The covariance value is positive ( $\text{Cov}(x, y) = 15$ ).

Interpretation: There is a positive relationship between treatment cost and length of hospital stay. As the hospital stay increases, the treatment cost tends to increase as well.

### 3. Outlier Detection using Z-Score (Patient Recovery Time) (10 Marks)

Recovery time (days) for patients after surgery is recorded as:

8, 9, 10, 10, 11, 12, 12, 13, 14, 15, 35

Perform the following:

1. Compute the mean.
2. Compute the standard deviation.
3. Calculate the z-score for each observation.
4. Identify the outlier(s) using the criterion .

ANSWER :-

**Outlier Detection using Z-Score :-**

Given recovery times (in days) for  $n = 11$  patients:

8, 9, 10, 10, 11, 12, 12, 13, 14, 15, 35

1. Compute the mean ( $\mu$ )

$$\mu = \frac{\sum x_i}{n} = \frac{8 + 9 + 10 + 10 + 11 + 12 + 12 + 13 + 14 + 15 + 35}{11} = \frac{149}{11} \approx 13.55$$

**Compute the standard deviation ( $\sigma$ ):-**

Using the population standard deviation formula  $\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$ :

$$\begin{aligned}\sum (x_i - \mu)^2 &= (8 - 13.55)^2 + (9 - 13.55)^2 + \dots + (35 - 13.55)^2 \\ &= 30.80 + 20.70 + 12.60 + 12.60 + 6.50 + 2.40 + 2.40 + 0.20 + 0.20 + 2.10 + 460.10 \\ &= 550.60\end{aligned}$$

$$\sigma = \sqrt{\frac{550.60}{11}} = \sqrt{50.05} \approx 7.07$$

(If sample standard deviation  $s$  with  $n - 1 = 10$  is used:  $s = \sqrt{\frac{550.60}{10}} \approx 7.42$ )

**Calculate the z-score for each observation ( $z = \frac{x_i - \mu}{\sigma}$ )**

(Using  $\mu = 13.55$  and  $\sigma = 7.07$ ):

| <b>xi</b> | <b><math>x_i - \mu</math></b> | <b>Z-Score (z)</b> |
|-----------|-------------------------------|--------------------|
| 8         | -5.55                         | -0.79              |
| 9         | -4.55                         | -0.64              |
| 10        | -3.55                         | -0.50              |
| 10        | -3.55                         | -0.50              |
| 11        | -2.55                         | -0.36              |
| 12        | -1.55                         | -0.22              |
| 12        | -1.55                         | -0.22              |
| 13        | -0.55                         | -0.08              |
| 14        | 0.45                          | +0.06              |
| 15        | 1.45                          | +0.21              |
| 35        | 21.45                         | +3.03              |

**Identify the outlier(s) using  $|z| > 3$**

- For  $x = 35$ ,  $|z| = 3.03 > 3$ .
- For all other values,  $|z| < 3$ .

Outlier identified: 35

#### 4. Data Discretization using Equal-Width Binning (Blood Glucose Levels) (5 Marks)

A clinic records fasting blood glucose levels (mg/dL).

Sorted Data:

75, 82, 88, 92, 95, 99, 104, 110, 118, 122, 130, 145

Partition the data into 3 equal-width bins.

Replace each observation by the nearest bin boundary.

Write the transformed dataset.

ANSWER :-

Data Discretization using Equal-Width Binning (5 Marks)

Sorted Data ( $N = 12$ ):

75, 82, 88, 92, 95, 99, 104, 110, 118, 122, 130, 145

**Step 1: Calculate Bin Width**

$$\text{Bin Width} = \frac{\text{Max} - \text{Min}}{k} = \frac{145 - 75}{3} = \frac{70}{3} \approx 23.33$$

**Step 2: Define Bin Boundaries**

- Bin 1:  $[75, 75 + 23.33) = [75, 98.33)$ 
  - Boundaries: 75 and 98.33
- Bin 2:  $[98.33, 98.33 + 23.33) = [98.33, 121.66)$ 
  - Boundaries: 98.33 and 121.66
- Bin 3:  $[121.66, 145]$ 
  - Boundaries: 121.66 and 145

**Step 3: Partition Data into Bins**

- Bin 1: 75, 82, 88, 92, 95
- Bin 2: 99, 104, 110, 118
- Bin 3: 122, 130, 145

#### Step 4: Replace each observation with nearest bin boundary

- Bin 1 (Boundaries: 75 and 98.33):
  - 75 → 75
  - 82 → 75 (*distance to 75 is 7, to 98.33 is 16.33*)
  - 88 → 98.33 (*distance to 75 is 13, to 98.33 is 10.33*)
  - 92 → 98.33
  - 95 → 98.33
- Bin 2 (Boundaries: 98.33 and 121.66):
  - 99 → 98.33
  - 104 → 98.33
  - 110 → 121.66
  - 118 → 121.66
- Bin 3 (Boundaries: 121.66 and 145):
  - 122 → 121.66
  - 130 → 121.66
  - 145 → 145

#### Transformed Dataset

**75, 75, 98.33, 98.33, 98.33, 98.33, 98.33, 121.66, 121.66, 121.66, 121.66, 145**



### 5. Standardization of Insurance Claim Amounts (10 Marks)

An insurance company records claim amounts (₹ thousand):

12, 15, 18, 20, 22, 25, 28, 30, 35, 40

Perform the following:

1. Compute the mean.
2. Compute the standard deviation.
3. Calculate the z-score for every observation.
4. Identify claims that are above one standard deviation from the mean.

ANSWER :-

Given claim amounts ( $n = 10$ ):

12, 15, 18, 20, 22, 25, 28, 30, 35, 40

Compute the mean ( $\mu$ )

$$\mu = \frac{\sum x_i}{n} = \frac{12 + 15 + 18 + 20 + 22 + 25 + 28 + 30 + 35 + 40}{10} = \frac{245}{10} = 24.5$$

2. Compute the standard deviation ( $\sigma$ )

Using population standard deviation:

$$\begin{aligned}\sum (x_i - \mu)^2 &= (12 - 24.5)^2 + (15 - 24.5)^2 + \dots + (40 - 24.5)^2 \\ &= 156.25 + 90.25 + 42.25 + 20.25 + 6.25 + 0.25 + 12.25 + 30.25 + 110.25 + 240.25 \\ &= 708.5\end{aligned}$$

$$\sigma = \sqrt{\frac{708.5}{10}} = \sqrt{70.85} \approx 8.42$$

(If sample standard deviation  $s$  with  $n - 1 = 9$  is used:  $s = \sqrt{\frac{708.5}{9}} \approx 8.87$ )

3. Calculate the z-score for every observation ( $z = \frac{x_i - \mu}{\sigma}$ )

(Using  $\mu = 24.5$  and  $\sigma = 8.42$ ):

| Claim Amount ( $x_i$ ) | $x_i - \mu$ | Z-score (z) |
|------------------------|-------------|-------------|
| 12                     | -12.5       | -1.45       |
| 15                     | -9.5        | -1.13       |
| 18                     | -6.5        | -0.77       |
| 20                     | -4.5        | -0.53       |
| 22                     | -2.5        | -0.30       |
| 25                     | 0.5         | +0.06       |
| 28                     | 3.5         | +0.42       |
| 30                     | 5.5         | +0.65       |
| 35                     | 10.5        | +1.25       |
| 40                     | 15.5        | +1.84a      |

**Identify claims above one standard deviation from the mean ( $z > 1$ )**

Threshold value =  $\mu + 1\sigma = 24.5 + 8.42 = 32.92$

- $x = 35$  ( $z = 1.25 > 1$ )
- $x = 40$  ( $z = 1.84 > 1$ )

Claims above 1 standard deviation: ₹35 thousand and ₹40 thousand